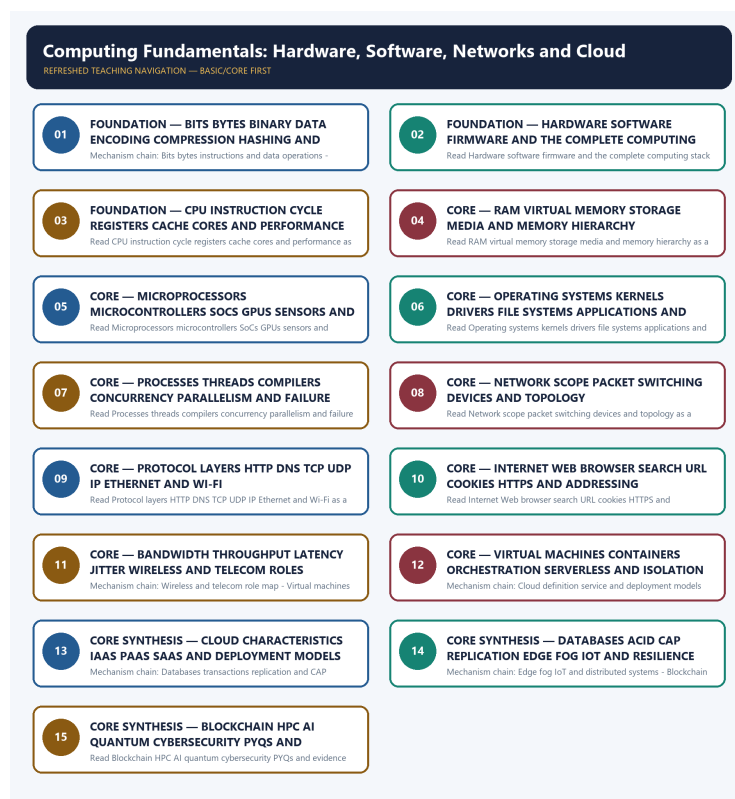


SOLVED PRACTICE WORKBOOK

Computing Fundamentals: Hardware, Software, Networks and Cloud - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Bits bytes instructions and data operations?

A. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. B. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. C. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. D. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM.

Answer: A. Explanation: A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Bits bytes instructions and data operations?

A. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. B. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. C. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. D. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM.

Answer: B. Explanation: A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Bits bytes instructions and data operations without changing its institution, unit or status?

A. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. B. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. C. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. D. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM.

Answer: C. Explanation: A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Bits bytes instructions and data operations?

A. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. B. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. C. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. D. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable.

Answer: D. Explanation: A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Computing stack boundary?

A. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. B. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. C. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. D. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking.

Answer: A. Explanation: A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Computing stack boundary?

A. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. B. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. C. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. D. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM.

Answer: B. Explanation: A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Computing stack boundary without changing its institution, unit or status?

A. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. B. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. C. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. D. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores.

Answer: C. Explanation: A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Computing stack boundary?

A. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. B. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. C. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. D. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical.

Answer: D. Explanation: A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies CPU instruction cycle and performance?

A. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. B. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. C. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. D. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website.

Answer: A. Explanation: A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. The other options change the scientific category, institutional owner, measurement, mission run or operating status.

Q10. Which option preserves the technical boundary of CPU instruction cycle and performance?

A. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. B. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. C. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. D. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website.

Answer: B. Explanation: A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. The other options change the scientific category, institutional owner, measurement, mission run or operating status.

Q11. Which statement uses CPU instruction cycle and performance without changing its institution, unit or status?

A. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. B. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. C. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. D. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security.

Answer: C. Explanation: A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about CPU instruction cycle and performance?

A. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. B. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. C. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. D. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking.

Answer: D. Explanation: A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Memory storage and volatility hierarchy?

A. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. B. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. C. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. D. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website.

Answer: A. Explanation: Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Memory storage and volatility hierarchy?

A. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. B. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. C. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. D. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website.

Answer: B. Explanation: Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Memory storage and volatility hierarchy without changing its institution, unit or status?

A. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. B. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. C. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. D. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores.

Answer: C. Explanation: Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Memory storage and volatility hierarchy?

A. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. B. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. C. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. D. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM.

Answer: D. Explanation: Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Processor forms and embedded systems?

A. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. B. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. C. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. D. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security.

Answer: A. Explanation: A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Processor forms and embedded systems?

A. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. B. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. C. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. D. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role.

Answer: B. Explanation: A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Processor forms and embedded systems without changing its institution, unit or status?

A. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. B. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. C. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. D. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security.

Answer: C. Explanation: A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Processor forms and embedded systems?

A. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. B. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. C. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. D. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world.

Answer: D. Explanation: A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Software OS and application layers?

A. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. B. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. C. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. D. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores.

Answer: A. Explanation: System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Software OS and application layers?

A. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. B. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. C. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. D. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role.

Answer: B. Explanation: System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Software OS and application layers without changing its institution, unit or status?

A. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. B. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. C. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. D. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true.

Answer: C. Explanation: System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Software OS and application layers?

A. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. B. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. C. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. D. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website.

Answer: D. Explanation: System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Processes threads concurrency and execution?

A. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. B. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. C. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. D. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security.

Answer: A. Explanation: A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Processes threads concurrency and execution?

A. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. B. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. C. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. D. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration.

Answer: B. Explanation: A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Processes threads concurrency and execution without changing its institution, unit or status?

A. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. B. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. C. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. D. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration.

Answer: C. Explanation: A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Processes threads concurrency and execution?

A. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. B. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. C. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. D. A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores.

Answer: D. Explanation: A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Networks packets and devices?

A. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. B. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. C. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. D. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration.

Answer: A. Explanation: PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Networks packets and devices?

A. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. B. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. C. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. D. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer.

Answer: B. Explanation: PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Networks packets and devices without changing its institution, unit or status?

A. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. B. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. C. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. D. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration.

Answer: C. Explanation: PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Networks packets and devices?

A. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. B. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. C. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. D. PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security.

Answer: D. Explanation: PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Layered protocol roles?

A. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. B. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. C. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. D. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true.

Answer: A. Explanation: Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Layered protocol roles?

A. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. B. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. C. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. D. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless.

Answer: B. Explanation: Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Layered protocol roles without changing its institution, unit or status?

A. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. B. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. C. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. D. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless.

Answer: C. Explanation: Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Layered protocol roles?

A. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. B. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. C. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. D. Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role.

Answer: D. Explanation: Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Internet web browser and cookie boundary?

A. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. B. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. C. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. D. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration.

Answer: A. Explanation: The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Internet web browser and cookie boundary?

A. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. B. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. C. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. D. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer.

Answer: B. Explanation: The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Internet web browser and cookie boundary without changing its institution, unit or status?

A. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. B. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. C. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. D. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself.

Answer: C. Explanation: The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Internet web browser and cookie boundary?

A. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. B. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. C. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. D. The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true.

Answer: D. Explanation: The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Addressing routing and configuration?

A. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. B. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. C. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. D. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless.

Answer: A. Explanation: An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Addressing routing and configuration?

A. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. B. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. C. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. D. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself.

Answer: B. Explanation: An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Addressing routing and configuration without changing its institution, unit or status?

A. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. B. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. C. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. D. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan.

Answer: C. Explanation: An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Addressing routing and configuration?

A. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. B. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. C. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. D. An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration.

Answer: D. Explanation: An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Network performance measures?

A. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. B. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. C. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. D. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself.

Answer: A. Explanation: Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Network performance measures?

A. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. B. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. C. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. D. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself.

Answer: B. Explanation: Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Network performance measures without changing its institution, unit or status?

A. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. B. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. C. Bandwidth is theoretical or provisioned capacity, throughput is achieved

data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. D. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system.

Answer: C. Explanation: Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Network performance measures?

A. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. B. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. C. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. D. Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer.

Answer: D. Explanation: Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Wireless and telecom role map?

A. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. B. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. C. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. D. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself.

Answer: A. Explanation: LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports

centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Wireless and telecom role map?

A. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. B. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. C. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. D. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan.

Answer: B. Explanation: LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Wireless and telecom role map without changing its institution, unit or status?

A. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. B. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. C. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. D. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system.

Answer: C. Explanation: LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Wireless and telecom role map?

A. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. B. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. C. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. D. LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless.

Answer: D. Explanation: LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Virtual machines containers and orchestration?

A. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. B. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. C. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. D. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan.

Answer: A. Explanation: A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Virtual machines containers and orchestration?

A. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. B. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. C. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. D. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service.

Answer: B. Explanation: A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Virtual machines containers and orchestration without changing its institution, unit or status?

A. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. B. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. C. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. D. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems.

Answer: C. Explanation: A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Virtual machines containers and orchestration?

A. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. B. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. C. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. D. A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs.

Answer: D. Explanation: A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Cloud definition service and deployment models?

A. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. B. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. C. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. D. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan.

Answer: A. Explanation: NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Cloud definition service and deployment models?

A. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. B. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. C. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. D. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system.

Answer: B. Explanation: NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Cloud definition service and deployment models without changing its institution, unit or status?

A. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. B. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. C. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. D. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs.

Answer: C. Explanation: NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Cloud definition service and deployment models?

A. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. B. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. C. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. D. NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself.

Answer: D. Explanation: NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Databases transactions replication and CAP?

A. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. B. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. C. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. D. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems.

Answer: A. Explanation: A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Databases transactions replication and CAP?

A. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. B. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. C. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. D. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems.

Answer: B. Explanation: A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Databases transactions replication and CAP without changing its institution, unit or status?

A. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. B. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. C. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. D. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems.

Answer: C. Explanation: A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Databases transactions replication and CAP?

A. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. B. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. C. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. D. A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan.

Answer: D. Explanation: A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Edge fog IoT and distributed systems?

A. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. B. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. C. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. D. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service.

Answer: A. Explanation: Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Edge fog IoT and distributed systems?

A. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. B. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. C. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. D. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable.

Answer: B. Explanation: Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Edge fog IoT and distributed systems without changing its institution, unit or status?

A. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. B. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. C. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. D. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical.

Answer: C. Explanation: Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Edge fog IoT and distributed systems?

A. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. B. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. C. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. D. Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system.

Answer: D. Explanation: Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Blockchain Web3 NFT and oracle boundary?

A. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. B. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. C. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. D. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems.

Answer: A. Explanation: A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Blockchain Web3 NFT and oracle boundary?

A. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. B. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. C. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. D. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable.

Answer: B. Explanation: A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Blockchain Web3 NFT and oracle boundary without changing its institution, unit or status?

A. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. B. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. C. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. D. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical.

Answer: C. Explanation: A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Blockchain Web3 NFT and oracle boundary?

A. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. B. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. C. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. D. A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service.

Answer: D. Explanation: A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies HPC AI quantum and semiconductor boundaries?

A. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. B. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. C. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. D. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable.

Answer: A. Explanation: HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of HPC AI quantum and semiconductor boundaries?

A. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. B. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. C. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. D. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking.

Answer: B. Explanation: HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses HPC AI quantum and semiconductor boundaries without changing its institution, unit or status?

A. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. B. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. C. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. D. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM.

Answer: C. Explanation: HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about HPC AI quantum and semiconductor boundaries?

A. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. B. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. C. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. D. HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems.

Answer: D. Explanation: HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Cybersecurity resilience and evidence firewall?

A. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. B. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. C. A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. D. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical.

Answer: A. Explanation: Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Cybersecurity resilience and evidence firewall?

A. A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. B. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. C. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. D. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM.

Answer: B. Explanation: Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Cybersecurity resilience and evidence firewall without changing its institution, unit or status?

A. A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. B. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. C. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. D. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world.

Answer: C. Explanation: Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Cybersecurity resilience and evidence firewall?

A. Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. B. A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. C. System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. D. Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs.

Answer: D. Explanation: Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Semantic-completeness coverage drills - Topic 25

Drill	Prompt	Minimum answer route	Fatal trap
A	Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM.		
B	A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world.		
C	System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website.		
D	Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs.		
E	Test traveller knowledge.	Kumarajiva, Faxian, Xuanzang and Yijing by route, purpose and date	neutral universal eyewitness
F	Explain Southeast Asian localization.	Funan/Champa -> Srivijaya/Sailendra -> local scripts, courts and monuments	Indian-built replicas
G	What can coins, ceramics and ships prove?	context -> circulation/use -> alternative explanation -> limit	coin/import equals rule
H	Demonstrate reciprocity.	foreign communities/motifs/technologies -> effects in India -> unequal regional exchange	one-way cultural export

PYQ status drill: all seven retained PYQs are adjacent or boundary-owned; none is routed directly to Topic 25.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route fifteen historical 2018-2022 objective demands and the direct 2026 blockchain demand to this owner, with several shared specialist routes. The recent metaverse, Majorana-chip and Kavach demands are used only for bounded computing support because their direct ledger owners are Economy, Topic 10 and Geography. Three representative cards preserve these ownership distinctions and no objective key.

PYQ DEMAND CARD 1 - 2018-2022 Prelims GS-I

Demand: Assess the routed foundations involving APIs, IoT, LTE and VoLTE, AR and VR, digital signatures, wearables, AI, VLC, blockchain, Web3, SaaS, qubits, short-range wireless systems and NFTs.

Status: Representative historical objective card covering fifteen routed or cross-routed demands; official keys are unavailable locally and no option or answer letter is supplied.

Model solution: Software OS and application layers: System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. **Layered protocol roles:** Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. **Wireless and telecom role map:** LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. **Cloud definition service and deployment models:** NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. **Edge fog IoT and distributed systems:** Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. **Blockchain Web3 NFT and oracle boundary:** A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **HPC AI quantum and semiconductor boundaries:** HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. **Cybersecurity resilience and evidence firewall:** Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 1 - 2018-2022 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Software OS and application layers: System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. **Layered protocol roles:** Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role.

Wireless and telecom role map: LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless.

Cloud definition service and deployment models: NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself.

Edge fog IoT and distributed systems: Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system.

Blockchain Web3 NFT and oracle boundary: A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service.

HPC AI quantum and semiconductor boundaries: HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems.

Cybersecurity resilience and evidence firewall: Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess the routed foundations involving APIs, IoT, LTE and VoLTE, AR and VR, digital signatures, wearables, AI, VLC, blockchain, Web3, SaaS, qubits, short-range wireless systems and NFTs. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Representative historical objective card covering fifteen routed or cross-routed demands; official keys are unavailable locally and no option or answer letter is supplied. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Software OS and application layers: System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages

privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. **Layered protocol roles:** Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role.

Wireless and telecom role map: LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. **Cloud definition service and deployment models:** NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. **Edge fog IoT and distributed systems:** Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. **Blockchain Web3 NFT and oracle boundary:** A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **HPC AI quantum and semiconductor boundaries:** HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. **Cybersecurity resilience and evidence firewall:** Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 1 - 2018-2022 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 2 - 2024-2025 Prelims GS-I

Demand: Apply the computing-layer distinctions needed for the metaverse, Majorana-chip and Kavach-RFID demands while retaining their specialist owners.

Status: Representative recent cross-owner card: Economy, Quantum Technology and Geography retain direct ownership; this topic supplies AR-VR, hardware-software, quantum-boundary and RFID mechanism only, without reproducing official Set-A keys.

Model solution: Processor forms and embedded systems: A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. **Wireless and telecom role map:** LTE is a packet-data cellular standard family and VoLTE carries

voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. **Blockchain Web3 NFT and oracle boundary:** A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **HPC AI quantum and semiconductor boundaries:** HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 2 - 2024-2025 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Processor forms and embedded systems: A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. **Wireless and telecom role map:** LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. **Blockchain Web3 NFT and oracle boundary:** A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **HPC AI quantum and semiconductor boundaries:** HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Apply the computing-layer distinctions needed for the metaverse, Majorana-chip and Kavach-RFID demands while retaining their specialist owners. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Processor forms and embedded systems: A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. **Wireless and telecom role map:** LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices,

NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. **Blockchain Web3 NFT and oracle boundary:** A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **HPC AI quantum and semiconductor boundaries:** HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 2 - 2024-2025 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 3 - 2026 Prelims GS-I

Demand: Assess blockchain database replication, practical immutability, stakeholder access and consortium participation models.

Status: Verified direct routed objective demand covering 2026 Q86; the locally held Set-A key is provisional, so no option, answer letter or inferred key is recorded.

Model solution: Databases transactions replication and CAP: A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. **Blockchain Web3 NFT and oracle boundary:** A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **Cybersecurity resilience and evidence firewall:** Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 3 - 2026 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Databases transactions replication and CAP: A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. **Blockchain Web3 NFT and oracle boundary:** A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested

decentralised-design umbrella rather than every future web service. **Cybersecurity resilience and evidence firewall:** Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess blockchain database replication, practical immutability, stakeholder access and consortium participation models. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified direct routed objective demand covering 2026 Q86; the locally held Set-A key is provisional, so no option, answer letter or inferred key is recorded. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Databases transactions replication and CAP: A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. **Blockchain Web3 NFT and oracle boundary:** A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **Cybersecurity resilience and evidence firewall:** Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 3 - 2026 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish data representation, hardware, software and processor-memory functions in a complete computing system. Answer in about 150 words.

Model thesis: Claim: Bits bytes instructions and data operations. **Named evidence/example:** A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution,

application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Computing stack boundary. **Named evidence/example:** A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CPU instruction cycle and performance. **Named evidence/example:** A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Memory storage and volatility hierarchy. **Named evidence/example:** Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Processor forms and embedded systems. **Named evidence/example:** A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable.
- A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical.
- A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking.
- Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM.
- A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world.

Qualified conclusion: Claim: Bits bytes instructions and data operations. **Named evidence/example:** A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Computing stack boundary. **Named evidence/example:** A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CPU instruction cycle and performance. **Named evidence/example:** A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Memory storage and volatility hierarchy. **Named evidence/example:** Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Processor forms and embedded systems. **Named evidence/example:** A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **answer** requires a direct position on "Distinguish data representation, hardware, software and processor-memory functions in a...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Bits bytes instructions and data operations. **Named evidence/example:** A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Computing stack boundary. **Named evidence/example:** A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date,

unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CPU instruction cycle and performance. **Named evidence/example:** A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Memory storage and volatility hierarchy. **Named evidence/example:** Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Processor forms and embedded systems. **Named evidence/example:** A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control,

and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Bits bytes instructions and data operations. **Named evidence/example:** A bit is a binary 0 or 1 and a byte normally contains eight bits; an instruction is an encoded processor operation, a program is an ordered instruction set and an algorithm is a finite language-independent method. Encoding changes representation, compression reduces size, hashing produces a fixed-size digest and encryption protects confidentiality with a key; these operations are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Computing stack boundary. **Named evidence/example:** A complete computing system links physical semiconductor devices, processor, memory, storage and input-output hardware to firmware, operating system, drivers, libraries, applications, data, networks and governance. Hardware executes and stores signals, software supplies instructions and rules, and a network connects systems; a product label may span layers without making the layers identical. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CPU instruction cycle and performance. **Named evidence/example:** A processor fetches, decodes and executes instructions; its control unit directs operations, arithmetic logic unit performs arithmetic and logic, registers hold immediate working values and cache reduces access delay to frequently used data. Clock rate, instructions per cycle, core count, cache, memory bandwidth, software, workload and thermal limits jointly shape performance, so gigahertz alone is not a ranking. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Memory storage and volatility hierarchy. **Named evidence/example:** Registers and cache are fastest and smallest, RAM is volatile working memory, while SSD, flash, HDD and optical or tape media provide non-volatile storage with different speed, cost and durability. Virtual memory maps process addresses to physical memory and storage to support isolation and a larger apparent address space; it does not create free physical RAM. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Processor forms and embedded systems. **Named evidence/example:** A microprocessor is a general-purpose CPU commonly paired with external memory and peripherals, a microcontroller integrates processor, memory and interfaces for embedded control, and a system-on-chip integrates multiple system components. GPUs or accelerators suit highly parallel workloads but do not universally replace CPUs; sensors measure, processors compute and actuators change the physical world. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Distinguish data representation, hardware, software and processor-memory functions in a...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain the role of operating systems, processes, threads and program-translation tools. Answer in about 150 words.

Model thesis: Claim: Software OS and application layers. **Named evidence/example:** System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Processes threads concurrency and execution. **Named evidence/example:** A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website.
- A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores.

Qualified conclusion: Claim: Software OS and application layers. **Named evidence/example:** System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Processes threads concurrency and execution. **Named evidence/example:** A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain the role of operating systems, processes, threads and program-translation tools....", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Software OS and application layers. **Named evidence/example:** System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Processes threads concurrency and execution. **Named evidence/example:** A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: **Claim:** Software OS and application layers. **Named evidence/example:** System software includes operating systems, drivers, runtimes and utilities, while application software performs user-facing tasks. The operating-system kernel manages privileged hardware and resources, a driver connects the OS to a device, a file system organises stored data, firmware controls low-level startup or device behaviour, and an API defines software interaction rather than a user-facing website. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Processes threads concurrency and execution. **Named evidence/example:** A process is a running program with resources and a thread is an execution path within it; concurrency means overlapping progress while parallelism means simultaneous execution. Compilers translate source before execution, interpreters translate or execute during runtime and assemblers convert assembly language; race conditions and deadlocks arise from coordination failures, not simply from using multiple cores. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain the role of operating systems, processes, threads and program-translation tools....", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse network devices, protocol layers, Internet-Web distinctions and performance measures. Answer in about 250 words.

Model thesis: **Claim:** Networks packets and devices. **Named evidence/example:** PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Layered protocol roles. **Named evidence/example:** Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Internet web browser and cookie boundary. **Named evidence/example:** The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Addressing routing and configuration. **Named evidence/example:** An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Network performance measures. **Named evidence/example:** Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security.
- Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role.
- The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true.

- An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration.
- Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer.

Qualified conclusion: Claim: Networks packets and devices. **Named evidence/example:** PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Layered protocol roles. **Named evidence/example:** Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Internet web browser and cookie boundary. **Named evidence/example:** The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Addressing routing and configuration. **Named evidence/example:** An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Network performance measures. **Named evidence/example:** Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **analyse** requires a direct position on "Analyse network devices, protocol layers, Internet-Web distinctions and performance measures....", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Networks packets and devices. **Named evidence/example:** PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment

status boundary. **Claim:** Layered protocol roles. **Named evidence/example:** Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Internet web browser and cookie boundary. **Named evidence/example:** The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Addressing routing and configuration. **Named evidence/example:** An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Network performance measures. **Named evidence/example:** Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses

versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration.

Analysis: Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

5. **Claim and named evidence:** Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Networks packets and devices. **Named evidence/example:** PAN, LAN, MAN and WAN classify typical network scope; packet switching divides data for transmission and reassembly. A switch forwards frames within a local network, a router forwards IP packets between networks, a modem converts signals for an access medium, an access point connects wireless clients to a LAN, a repeater regenerates signals and a firewall applies traffic rules; none alone supplies end-to-end security. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Layered protocol roles. **Named evidence/example:** Application protocols such as HTTP and DNS serve user or naming functions, transport protocols such as TCP and UDP carry application data, IP addresses and routes packets across networks, link technologies such as Ethernet or Wi-Fi deliver locally, and the physical layer carries electrical, optical or radio signals. Layering permits interoperability; a protocol at one layer does not perform every other layer's role. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Internet web browser and cookie boundary. **Named evidence/example:** The Internet is the global system of interconnected IP networks, while the World Wide Web is a service using HTTP or HTTPS over that infrastructure. A browser retrieves and renders content, a search engine indexes and searches it, a URL identifies a resource, DNS resolves names, and a cookie stores state or identifiers in a browser; HTTPS protects transport but does not certify that content is true. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Addressing routing and configuration. **Named evidence/example:** An IP address is a logical routing address while a MAC address operates at the local link layer; routing selects paths, DHCP supplies network configuration and NAT translates address information, commonly allowing private IPv4 devices to share a public address. IPv6 uses 128-bit addresses versus IPv4's 32 bits, but expanded addressing does not by itself guarantee security, privacy, speed or migration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Network performance measures. **Named evidence/example:** Bandwidth is theoretical or provisioned capacity, throughput is achieved data rate, latency is delay, jitter is delay variation and packet loss is failed delivery. High bandwidth can coexist with high latency, while voice, gaming, remote control and industrial systems may be more sensitive to delay and jitter than bulk file transfer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Analyse network devices, protocol layers, Internet-Web distinctions and performance measures....", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Compare wireless technologies, virtual machines, containers and cloud service and deployment models. Answer in about 250 words.

Model thesis: Claim: Wireless and telecom role map. **Named evidence/example:** LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Virtual machines containers and orchestration. **Named evidence/example:** A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cloud definition service and deployment models. **Named evidence/example:** NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless.
- A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs.
- NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself.

Qualified conclusion: Claim: Wireless and telecom role map. **Named evidence/example:** LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Virtual machines containers and orchestration. **Named evidence/example:** A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cloud definition service and deployment models. **Named evidence/example:** NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **compare** requires a direct position on "Compare wireless technologies, virtual machines, containers and cloud service and deployment...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Wireless and telecom role map. **Named evidence/example:** LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Virtual machines containers and orchestration. **Named evidence/example:** A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cloud definition service and deployment models. **Named evidence/example:** NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: **Claim:** Wireless and telecom role map. **Named evidence/example:** LTE is a packet-data cellular standard family and VoLTE carries voice over LTE; Wi-Fi usually provides local networking, Bluetooth connects personal devices, NFC supports centimetre-scale proximity exchange, RFID identifies tags through a reader, low-power mesh supports sensors and VLC transmits data with visible light. Range, wall penetration, power and topology are technology-specific rather than implied by the word wireless. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Virtual machines containers and orchestration. **Named evidence/example:** A virtual machine emulates hardware and runs a guest operating system, while a container isolates an application and dependencies while sharing the host kernel; orchestration schedules and manages many containers or services. Serverless or function services execute provider-managed code on demand, but convenience introduces cold-start, observability, portability and provider-dependence trade-offs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cloud definition service and deployment models. **Named evidence/example:** NIST cloud computing combines on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. IaaS supplies virtual compute, storage and networking, PaaS supplies a managed platform or runtime and SaaS supplies a ready application; public, private, community and hybrid describe deployment models. A data centre or virtual machine may enable cloud without being cloud by itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Compare wireless technologies, virtual machines, containers and cloud service and deployment...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Examine databases, distributed systems, edge computing, IoT and blockchain through reliability and truth-boundary concerns. Answer in about 300 words.

Model thesis: Claim: Databases transactions replication and CAP. **Named evidence/example:** A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Edge fog IoT and distributed systems. **Named evidence/example:** Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blockchain Web3 NFT and oracle boundary. **Named evidence/example:** A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan.
- Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system.
- A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service.

Qualified conclusion: Claim: Databases transactions replication and CAP. **Named evidence/example:** A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Edge fog IoT and distributed systems. **Named evidence/example:** Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. **Analysis:** This fixes the scientific

process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blockchain Web3 NFT and oracle boundary. **Named**

evidence/example: A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **examine** requires a direct position on "Examine databases, distributed systems, edge computing, IoT and blockchain through...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Databases transactions replication and CAP. **Named evidence/example:** A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Edge fog IoT and distributed systems. **Named**

evidence/example: Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blockchain Web3 NFT and oracle boundary. **Named**

evidence/example: A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Databases transactions replication and CAP. **Named evidence/example:** A DBMS stores and manages data; relational systems use tables, keys and constraints, while NoSQL covers varied non-relational models. ACID means atomicity, consistency, isolation and durability for transactions; replication supports availability or locality but is not backup, and during a network partition CAP frames a trade between strict consistency and availability rather than an all-time choose-two slogan. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Edge fog IoT and distributed systems. **Named evidence/example:** Cloud centralises resources, edge computing processes near the source and fog supplies intermediate distributed processing; the choice depends on latency, bandwidth, privacy, resilience and management. An IoT chain joins sensor or device, local processing, connectivity, platform or cloud, analytics and alert or actuator; connectivity alone does not make an isolated electronic object an IoT system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blockchain Web3 NFT and oracle boundary. **Named evidence/example:** A distributed ledger is replicated across nodes and a blockchain is one design using grouped, cryptographically linked records and consensus; public, private and consortium participation models differ. Immutability is practical rather than absolute, an oracle can introduce false external data, an NFT does not automatically transfer copyright or guarantee an asset, and Web3 is a contested decentralised-design umbrella rather than every future web service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Examine databases, distributed systems, edge computing, IoT and blockchain through...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Evaluate HPC, AI, semiconductor, quantum and cybersecurity capability as a layered national technology stack. Answer in about 300 words.

Model thesis: Claim: HPC AI quantum and semiconductor boundaries. **Named evidence/example:** HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. **Analysis:** This fixes the scientific process, system boundary, institution,

application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Cybersecurity resilience and evidence firewall. **Named evidence/example:** Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems.
- Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs.

Qualified conclusion: Claim: HPC AI quantum and semiconductor boundaries. **Named evidence/example:** HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Cybersecurity resilience and evidence firewall. **Named evidence/example:** Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **evaluate** requires a direct position on "Evaluate HPC, AI, semiconductor, quantum and cybersecurity capability as a layered national...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: HPC AI quantum and semiconductor boundaries. **Named evidence/example:** HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Cybersecurity resilience and evidence firewall. **Named evidence/example:** Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: HPC AI quantum and semiconductor boundaries. **Named evidence/example:** HPC uses parallel systems for computationally intensive work, with performance dependent on processors, accelerators, interconnect, memory, storage, cooling, software and skills. The National Supercomputing Mission is jointly steered by DST and MeitY and implemented by C-DAC and IISc with National Knowledge Network access; AI is software capability, semiconductors are device foundations and quantum computers use qubits for selected problems rather than universally replacing classical systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Cybersecurity resilience and evidence firewall. **Named evidence/example:** Confidentiality, integrity and availability are separate security goals; authentication verifies identity and authorisation grants permissions, encryption protects confidentiality, hashing supports integrity, signatures support origin and integrity, and certificates bind keys to identities. A standard, encrypted channel, benchmark, demonstration, product announcement, deployed system, high availability, secure operation and verified public outcome are separate claims; computing also carries energy, cooling, material, e-waste and access costs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Evaluate HPC, AI, semiconductor, quantum and cybersecurity capability as a layered national...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.